

CS 4782: Final Project Instructions

This document contains all materials related to the final project presentation and evaluation:

1. Academic Poster Presentation Instructions
2. Poster Presentation Evaluation Rubric
3. 2-Page Project Summary Report
4. Final Project Deliverable (GitHub) Instructions

Academic Poster Presentation Instructions

Due: April 30 or May 5. Please click “Final Project Presentation Times” to see the date your group is presenting.

1. Introduction

This document outlines the instructions for preparing and submitting your academic poster presentation. Your poster should visually communicate your project’s motivation, methodology, results, and key takeaways. The poster will be presented during the class poster session, and each group will have an opportunity to discuss their work with instructors and peers. A sample poster for reference can be found [here](#).

2. Poster Format and Layout

Your poster should adhere to the following format:

- **Size:** Maximum Poster size : 36” x 24” - as a singular printed poster or multiple A4 sheets collated together.
- **Orientation:** Landscape format is preferred.
- **File Format:** Submit as a PDF on Github (refer to final deliverable instructions). Ensure fonts and images are embedded.

3. Poster Structure

Your poster should include the following sections:

1. **Title and Authors**
 - Title of your project
 - Names of all group members and institutional affiliation (You may utilize the headers and/or footer of your poster for this information to ensure optimal poster-space usage)
2. **Introduction / Background / Motivation**
 - Brief description of the problem being addressed.
 - Clear statement of the goal or hypothesis of your project.
 - Specific result you aimed to reproduce or investigate.
 - Context and motivation for your work.
 - Summary of the chosen paper and main contributions (if applicable).
3. **Methodology**
 - Overview of your approach, including models, datasets, and tools used.
 - Any modifications or design choices made.
 - Any interesting experiments your group came up with to add to the paper
4. **Results**
 - Present results in visual form: graphs, tables, charts, images.
 - Compare your outcomes with the original findings or expectations.
5. **Conclusion**

- Summarize key takeaways from your work.
6. **Future Work (Optional)**
- Mention possible directions for future exploration.
7. **References**
- Cite the original paper and any resources used.

4. Design Guidelines

- Use large fonts (minimum 24pt for body text, larger for headings).
- Use bullet points for clarity.
- Choose high-contrast colors and readable fonts.
- Use visuals effectively—avoid dense paragraphs.
- Ensure all visuals have captions and are referenced in the text.

5. Presentation Tips

- Prepare a 2–3 minute summary of your poster.
- Be ready to answer questions about your work.
- Practice clear and concise explanations.
- Interpret the results and highlight any patterns or discrepancies.
- Provide hypotheses for unexpected findings.

6. Peer Review of Posters

In addition to presenting your own work, each student will be assigned **TBD other posters** to review during the session. Naturally, you would only be reviewing on the day your group is **NOT** presenting. Your role as a reviewer is to engage with the presenters, ask thoughtful questions, and provide constructive feedback.

Reviewer Guidelines

- Read each assigned poster carefully and ask at least one question to the presenters.
- Fill out a short review form for each poster (details will be provided).
- Focus on clarity, rigor, originality, and visual design.
- Be respectful and supportive in your feedback.

Grading Note

Your participation as a reviewer will count toward your final grade. We expect thoughtful engagement with each poster and meaningful feedback.

7. Submission

- Submit your poster as a PDF on Github by the day you present
- Bring a printed version of your poster to class on the day of the session.
- Ensure your group is present during the entire poster session.

Poster Presentation Evaluation Rubric

Instructions

The following scoring scale and rubric will be used by all reviewers (staff/student) to assess the quality of the presentation. Each criterion is rated on a scale from 1 to 5:

Scoring Scale:

- **1 - Poor:** Not addressed or severely lacking.
- **2 - Limited:** Present but unclear or incomplete.
- **3 - Good:** Meets expectations.
- **4 - Very Good:** Clear, well-executed, and above average.
- **5 - Amazing:** Exceptionally clear, creative, and insightful; exceeds expectations.

Rubric:

ID	Evaluation Criteria	1	2	3	4	5
	Content					
A	Clarity of Motivation (Problem and context clearly introduced and made interesting)					
B	Description of Method (Approach is clearly summarized)					
C	Results and Evaluation (Outcomes are presented clearly and interpreted)					
D	Discussion and Insight (Challenges, analysis, and future work addressed)					
	Presentation Design					
E	Visual Clarity (Layout, readability, and visual balance)					
F	Use of Visuals (Graphs, images, or diagrams support the content)					
G	Text Economy (Minimal text, effective use of space)					
	Delivery and Engagement					
H	Verbal Summary (Clear 2–3 minute explanation)					
I	Engagement with Reviewers (Responsive, thoughtful discussion)					
	Initiative and Creativity					
J	Independent Exploration (Tried interesting additional experiments or ideas beyond the original paper)					
K	Overall Impression (How interesting, engaging, and worthwhile the poster was to review)					
	Total Score (out of 50):					

Just for Fun Awards!

Nominate 1 project for each:

Best Project/Presentation:

Funniest Poster/Presentation:

2-Page Project Summary Report

Due: May 12

Overview

Each group is required to submit a concise, well-organized 2-page summary of their project. This report should complement your poster presentation and clearly communicate the core motivation, approach, findings, and reflections from your work.

Submission Details

- **Length:** Maximum 2 pages (excluding images and references)
- **Format:** PDF, using 11–12pt font, single-spaced
- **Submission Platform:** GradeScope and GitHub
- **File Name:** `group_topic_2page_report.pdf`

Required Sections

1. Introduction

- Briefly introduce your project:
 - State the problem and motivation.
 - Include the title and authors of the paper (if reproducing one).
 - Summarize the primary method and contributions of the paper.

2. Chosen Result

- Identify the specific result you aimed to reproduce and its significance in the context of the paper's main contribution(s).
- Include the relevant figure, table, or equation reference from the original paper.
- State why it was chosen and its importance in the context of the paper or your goals.

3. Methodology

- Describe your re-implementation approach, including the model architecture, datasets, evaluation metrics, and any modifications made to the original methodology.

4. Results & Analysis

- Present your re-implementation results and compare them to the original paper's findings.
- Discuss any discrepancies or challenges encountered during the re-implementation process.
- Provide an analysis of your results in the context of the paper's main contribution(s) and the broader research area.
- **Note:** We are looking for a reasonable re-implementation of the method and a clear discussion of your results. A failure to match the reported results could happen for any number of reasons that would not negatively impact your grade. It is acceptable, for instance, to run smaller-scale experiments if you initially under-estimated the required resources for your selected result. It's also possible that the authors left out some detail that is necessary to match their performance.

5. Reflections

- Share lessons learned during the project.
- Summarize the key takeaways from your re-implementation effort and the lessons learned.
- Discuss potential future directions or extensions based on your findings and the paper's implications.

6. References

- Include full citations for the original paper and any tools, datasets, or frameworks used.

Tips for Writing

- Report should have more emphasis on the section 3 (Methodology), 4 (Results and Analysis) and 5 (Reflections). Use the available 2-page limit efficiently.
- Focus on clarity and conciseness — aim to inform a peer who hasn't seen your poster.
- Use bullet points, diagrams, or figures if helpful (but ensure they fit within the 2 pages).
- Avoid excessive jargon; explain important terms and assumptions.

Grading

There is not a specific grading rubric for the project report. However, we will look for the following factors:

- How much thought and consideration did you put into the problem setting? Is the problem setting sufficiently challenging or interesting to explore?
- How much thought and consideration did you put into your approach? Did you correctly identify the previous approaches and their limitations? Did you attempt to address these limitations through your work? Did you note key obstacles in implementing your approach? Even if your results were not successful, did you try enough things and reflect on why they did not work? What would you have done differently given more time and resources?
- How much thought and consideration did you put into your investigation? Did you use reasonable performance metrics? Did you provide insight and visualize key aspects of your approach (using tables, figures, toy examples)? For any theoretical components of the project, have you identified the crucial assumptions and limitations?
- Generally speaking the more iterations we see, the better the project quality.
- When grading we will also take into consideration the group sizes.

Final Project Deliverable (GitHub) Instructions

Due: May 12

1. Introduction

This document outlines the instructions for the final project deliverable for CS 4782. Each group should create a GitHub repository containing their re-implementation code, a detailed README, and any necessary data or resources. The README should provide a comprehensive overview of the project, including the chosen paper, the specific result reproduced, instructions for running the code, and an analysis of the re-implementation results in the context of the original paper's findings.

2. GitHub Repository Structure

Your GitHub repository should have the following structure:

- **README.md**: A detailed README file with the sections described below.
- **code/**: A directory containing your re-implementation code, along with any necessary configuration files or scripts.
- **data/**: A directory containing the datasets used for training and evaluation, or a README with instruction on how to obtain the dataset.
- **results/**: A directory containing the results of your re-implementation, including any generated figures, tables, or log files.
- **poster/**: A directory containing a PDF of the poster used for your in-class presentations.
- **report/**: A directory containing a PDF of the final report submitted.
- **LICENSE**: A file specifying the license under which your code is released (e.g., MIT, Apache 2.0).
- **.gitignore**: A file specifying files or directories that should be ignored by Git.

3. README Contents

The purpose of the README is to provide a TLDR snapshot of your work targeted towards anyone landing on your GitHub Repo. This is NOT A report! You can use the sample README here for reference.

Note that most of the content can be copied over concisely from your final report. Limit each section to 1-2 lines/figures. Your README should include the following sections:

1. Introduction

- Purpose of this Git repo (mention how this is a project that attempts to re-implement your paper of choice)
- Introduce the paper chosen and its main contribution.

2. Chosen Result

- Identify the specific result you aimed to reproduce and its significance in the context of the paper's main contribution(s).
- Include the relevant figure, table, or equation reference from the original paper.

3. GitHub Contents

- Make a brief note about the content structure of your project.

4. Re-implementation Details

- Describe your approach to re-implementation or experimentation.
- Include key details about models, datasets, tools, and evaluation metrics.
- Mention any challenges or modifications made to the original approach.

5. Reproduction Steps

As **meta** as this section is, it essentially documents steps someone would need to follow to implement your GitHub repo in a local environment.

- Describe "how someone using your GitHub can re-implement your re-implementation?"
- Provide instructions for running your code, including any dependencies, required libraries, and command-line arguments.
- Specify the computational resources (e.g., GPU) needed to reproduce your results.

6. Results/Insights

- Present your re-implementation results as a comparison to the original paper's findings. Describes "what can someone expect as the end-result of using your GitHub repo?"

7. Conclusion

- Summarize the key takeaways from your re-implementation effort and the lessons learned.

8. References

- Include a list of references, including the original paper and any additional resources used in your re-implementation.

9. Acknowledgements

- Recognition goes a long way in setting up the context of your work. Your acknowledgements also act as an indirect validation about the quality of the work. For e.g., having done this project as part of coursework is a sign that the work was potentially peer-reviewed or graded - i.e. added authenticity.

4. Submission

- Create a public GitHub repository for your project and add all group members as collaborators.
- Ensure that your repository is well-organized, with clear documentation and instructions for reproducing your results.
- Submit the link to your GitHub repository through Gradescope by May 12.